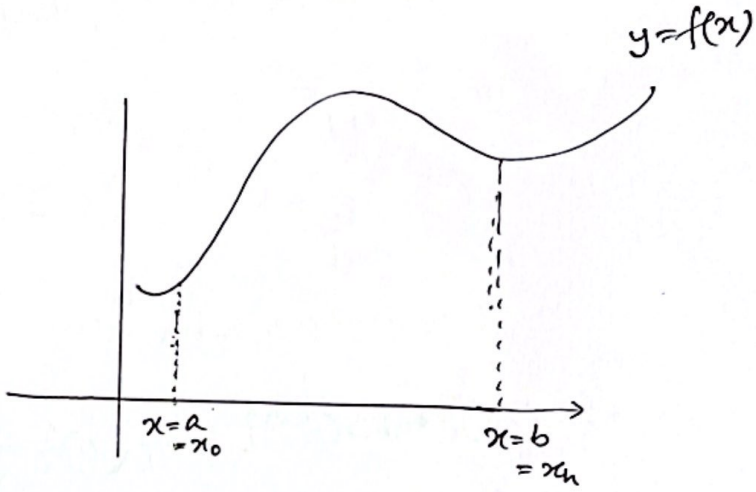
Lecture: 11

APPROXIMATE the area using a finite number of RECTANGLES

Given a curve : $y = f(x)$



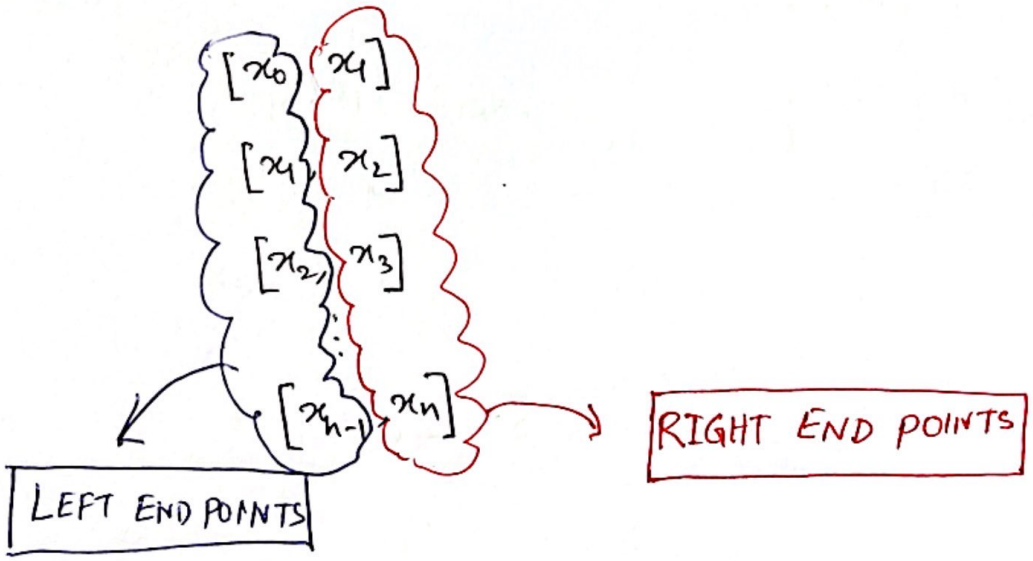
Suppose we use "n" rectangles.

Then each rectangle has a width we'll call Δx.

$$\Delta x = \frac{b-a}{n} = \frac{x_n - a}{n}$$

$$x_n = a + n \cdot \Delta x$$

We have n "sub-intervals" between $x = a$ & $x = b$.



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Here I'll use the RIGHT ENDPOINTS.

In each sub-interval we measure the AREA of a RECTANGLE:

$$A = \text{HEIGHT} \times \text{WIDTH}$$

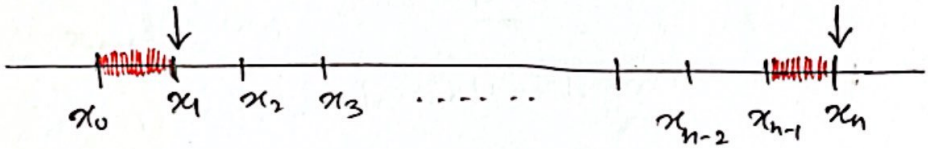
$$A_i = f(x_i) \cdot \Delta x$$

When using the RIGHT ENDPOINTS, the expression:

$$A \approx \sum_{i=1}^{n} f(x_i) \Delta x = f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x + \cdots + f(x_n)\Delta x$$

is called a RIGHT RIEMANN Sum: R_n

Here $\boxed{x_i = a + i\,\Delta x}$



x_1 is the RIGHT ENDPOINT of the 1st interval

x_2 " " " " " 2nd interval

x_3 " " " " " 3rd "

⋮

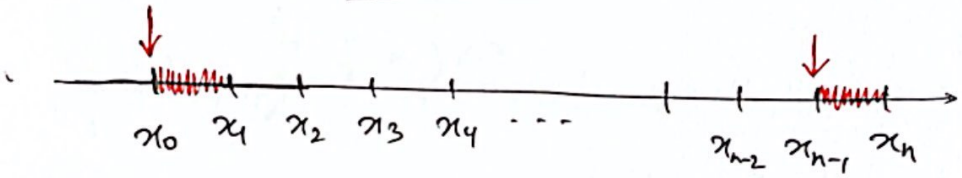
x_n " " " " " nth "

When using the LEFT END POINTS, the expression:

$$A \approx \sum_{i=0}^{n-1} f(x_i)\Delta x = f(x_0)\Delta x + f(x_1)\Delta x + \cdots + f(x_{n-1})\Delta x$$
$$= \sum_{i=1}^{n} f(x_i)\Delta x$$

is called a Left RIEMANN Sum $= L_n$

Here $\boxed{x_i = a + (i-1)\Delta x}$



x_0 is the LEFT END POINT of the 1st interval

x_1 - - - - - - - - - - 2nd "

x_2 — — 3rd "

⋮ ⋮ nth "

x_{n-1} — —

Some Important finite sum:

1. $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

2. $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

3. $\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$.

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Definition: Let $f(x)$ be a function that is defined on a finite interval $[a, b]$.

Let $\Delta x = \frac{b-a}{n}$ and $x_i = a + i \cdot \Delta x$.

then the Riemann sum

$$R_n = \sum_{i=1}^{n} f(x_i)\,\Delta x$$

Riemann proved that the Riemann Sum Approximation becomes EXACT as the value of n gets LARGE.
we do this by letting $n \to \infty$ and define:

$$\lim_{n\to\infty} \left(\sum_{i=1}^{n} f(x_i)\Delta x \right) = \int_a^b f(x)\,dx = A$$

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Problem: Given that $f(x) = \frac{x^2}{4} + 8$ on $[0, 3]$. Find Riemann sum R_n and also find $\lim_{n\to\infty} R_n$.

Sol^n.

Riemann Sum,

$$R_n = \sum_{i=1}^{n} f(x_i)\Delta x \quad ; \text{ here } \Delta x = \frac{3-0}{n} = \frac{3}{n}$$

$$x_i = a + i\Delta x = 0 + i\frac{3}{n} = \frac{3i}{n}$$

$$= \sum_{i=1}^{n} f\left(\frac{3i}{n}\right)\Delta x$$

$$= \sum_{i=1}^{n} \left[\frac{\left(\frac{3i}{n}\right)^2}{4} + 8\right]\cdot\frac{3}{n}$$

$$= \sum_{i=1}^{n} \frac{3}{n}\left[\left(\frac{3}{n}\right)^2\cdot\frac{1}{4}i^2 + 8\right]$$

$$= \left(\frac{3}{n}\right)^3\cdot\frac{1}{4}\sum_{i=1}^{n} i^2 + \frac{3}{n}\sum_{i=1}^{n} 8$$

$$= \frac{1}{4}\left(\frac{3}{n}\right)^3 \frac{n(n+1)(2n+1)}{6} + \frac{3}{n}\cdot 8n$$

$$= \frac{27}{24}\frac{1}{n^3} n^3\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right) + 24$$

$$= \frac{27}{24}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right) + 24$$

Now, $$\lim_{n\to\infty} R_n = \lim_{n\to\infty} \frac{27}{24}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right) + 24$$

$$= \frac{27}{24}\cdot(1+0)(2+0) + 24$$

$$= \frac{27}{24}\cdot 1\cdot 2 + 24$$

i.e. $$\int_0^3 f(x)\,dx = \lim_{n\to\infty} R_n = \frac{27}{12}\cdot\frac{9}{4} + 24 = \frac{105}{4}$$

※

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Select subinterval

1. Left endpoint: $x_i = a+(i-1)\Delta x$
2. Right endpoint: $x_i = a+i\Delta x$
3. Mid point : $x_i = a+(i-\frac{1}{2})\Delta x$

Extra Problems

Book: Calculus : Early Transcendental By H. Anton

PAGE: 351 :

PROBLEMS: 35-48

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Indefinite Integral

Lecture 2

Recap

We already know the antiderivative of a function f.

The process of finding antiderivetives is called antidifferentiation or integration.

If $F(x)$ be the antiderivative of $f(x)$ then,

$$\int f(x)\,dx = F(x) + C$$

Example: Evaluate the following indefinite integral.

$$\int x^4 + 3x - 9\ dx$$

Sol^n:

$$\int (x^4 + 3x - 9)\,dx = \frac{x^5}{5} + \frac{3}{2}x^2 - 9x + C$$

here C is integral constant. □

Integration Formulas

page 324 : Table 5.2.1 Book. Calculus: Early Transcendents — H. Anton

e.g.

$$\int dx = x + C$$

$$\int x^r\,dx = \frac{x^{r+1}}{r+1} + C \quad r \neq -1$$

$$\int \cos x\,dx = \sin x + C$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \sec^2 x\,dx = \tan x + C$$

$$\int \csc^2 x\,dx = -\cot x + C$$

and so on.

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Properties

Suppose that $F(x)$ and $G(x)$ are antiderivatives of $f(x)$ and $g(x)$ respectively, and a is a constant. Then:

(a) $\int af(x)\,dx = aF(x) + C$

(b) $\int [f(x) \pm g(x)]\,dx = F(x) \pm G(x) + C$

Example: $\int (3x^6 - 2x^2 + 7x)\,dx = 3\int x^6\,dx - 2\int x^2\,dx + 7\int x\,dx$

$$= 3\,\frac{x^{6+1}}{6+1} - 2\,\frac{x^{2+1}}{2+1} + 7\,\frac{x^2}{2} + C$$

$$= \frac{3}{7}x^7 - \frac{2}{3}x^3 + \frac{7}{2}x^2 + C$$

Problem: Evaluate: 1. $\int \frac{\cos x}{\sin^2 x}\,dx$

Sol^n

1. $\int \frac{\cos x}{\sin^2 x}\,dx = \int \frac{1}{\sin x}\,\frac{\cos x}{\sin x}\,dx$

$$= \int \csc x \cot x\,dx = -\csc x + C$$

2. $\int \frac{x^2}{x^2+1}\,dx = \int \frac{x^2+1-1}{x^2+1}\,dx = \int \left(\frac{x^2+1}{x^2+1} - \frac{1}{x^2+1}\right)dx$

$$= \int \left(1 - \frac{1}{1+x^2}\right)dx$$

$$= x - \tan^{-1}x + C$$

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Exercise: TRY YOURSELF !!!

Evaluate the following integral:

1. $\int \frac{x^5+2x^2-1}{x^4}\,dx$

2. $\int (3\sin x - 2\sec^2 x)\,dx$

3. $\int \frac{\sin x}{\cos^2 x}\,dx$

4. $\int (1+\sin^2\theta \csc\theta)\,d\theta$

5. $\int \left[\frac{1}{2\sqrt{1-x^2}} - \frac{3}{1+x^2}\right] dx$

Not Now

6. $\int \sqrt{16-x^2}\,dx$

7. ~~If $\int 3x^2 dx$~~

7. If $f'(x) = x^4+3x-9$, what was $f(x)$?

8. $\int \sin\left(\frac{t}{2}\right)\cos\left(\frac{t}{2}\right) dt$

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